Geometric Langlands and the Arthur-Ramanujan conjecture

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Slides available at: `www.samraskin.net/talk.pdf`

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3. $\theta_p \in \mathbf{R}$, i.e., $|\tau(p)| \leqslant 2p^{\frac{11}{2}}$.

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One of the primary goals of the Langlands program is to "understand" the eigenvalues of Hecke operators acting on automorphic forms more generally.

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For today, (unramified) automorphic forms for (G, X) are functions

$$f\colon \mathrm{Bun}_G(\mathbf{F}_q) \to \mathbf{e}$$

with finite support

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The relations for Hecke operators allow us to assign for each eigenfunction f and each $x \in X(\mathbf{F}_{q_x})$ an eigenvalue

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Roughly speaking, the *naive* Ramanujan conjecture says that for f cuspidal, $\mathbf{e} = \mathbf{C}$, the Hecke eigenvalues are unitary (lie in the maximal compact subgroup) up to a correction by a character (which is unnecessary if G is semi-simple, which we will sometimes assume below to simplify).

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This was the big achievement of Vincent's work on function fields (which also works with ramification, etc.).

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(continuous along the first factor and algebraic along the second) that is ***irreducible****. Moreover, the following diagram commutes:*

$$\begin{tikzcd}
\mathcal{W}_X \arrow[r, "g \mapsto (g, \begin{pmatrix} |g|^{\frac{1}{2}} & 0 \\ 0 & |g|^{-\frac{1}{2}} \end{pmatrix})"] \arrow[dr, "\sigma_f"'] & \mathcal{W}_X \times SL_2(\overline{\mathbf{Q}}_\ell) \arrow[d, "\sigma_f^{\mathrm{Arth}}"] \\
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2. The theorem applies a little more generally, to "fake discrete series." This allows us to apply it to the constant function 1 (which is not even finitely supported!). Here the corresponding Arthur parameter is $\mathcal{W}_X \times SL_2 \to SL_2 \to \check{G}$ via the principal SL_2.

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Thank you!